HARRISON PIELKE-LOMBARDO

Software Developer

pielkekid@gmail.com | 720 209 6249 | Denver, CO | https://www.github.com/tuh8888 | https://www.linkedin.com/in/tuh8888

EXPERIENCE

Senior Software Engineer
Sep 2022 - Present
Peruse Technology Inc (SOLD)
Remote

- Helped drive product development from zero to $2.1M annual revenue and over 70 B2B clients, culminating in a successful acquisition.
- Built critical features for a high-volume logistics document processing platform: automatic broker verification, document assignment workflows, custom rules engine, and the core load processing pipeline serving both UI and API customers.
- Executed a complex database migration from `Datomic` to `Postgres` while maintaining system reliability.
- Developed full-stack solutions using `Clojure` and `Python` backends (`REST` / `GraphQL`), `Typescript` frontend components, and `AWS` infrastructure managed with `terraform` . Trained and deployed LLM-based ML models using `PyTorch` .

Computational Biologist
Aug 2016 – Aug 2022
University of Colorado, Anschutz Medical Campus
Aurora, CO

- Managed interconnected computational projects written in `Python` , `Java` , and `Clojure` across 4+ teams through consistent communication and leadership and produced 6 publications, 4 dataset releases, and 5 package releases.

PROJECTS

Schematization of biological mechanisms from structural, semantic, and causal properties
2017 – 2022

- Conceptualized and tested algorithmic approaches in `Clojure` and `Python` for extending analogical reasoning of complex networks representing biological knowledge. Merged and harmonized large biomedical knowledge graphs represented in `RDF` , `Datomic` , and `SQL` formats.

Knowtator: Concept/relation annotation for Protégé
2016–2022

- Deployed a WYSIWYG `Java` plugin for ontology development and text annotation with interactive graph visualization. Used to create 100k+ concept and relation annotations. Evolved into a `Clojurescript` `web application` with `CI/CD` pipeline incorporating iterative user feedback.

Bootstrapped relation extraction using word embeddings and dependency paths
2016–2019

- Implemented a novel bootstrapping approach in `Clojure` for automated relation extraction, generating human-interpretable syntax patterns from dependency paths and word embeddings. Optimized with GPU-accelerated matrix operations, improving drug-target relation extraction precision by 10%.

PRESENTATIONS

IDEAL Invited Speaker
June 2024
The Institute for Data, Econometrics, Algorithms, and Learning
Chicago, IL

- Title: AI in the Transportation Sector: Challenges and Successes

EDUCATION

Master of Science in Computational Bioscience
2016 — 2022
University of Colorado, Anschutz Medical Campus
Aurora, CO

Bachelor of Science in Applied Mathematics
2013 – 2016
University of Colorado
Boulder, CO

- Minor in Computer Science
- Minor in Biochemistry

SKILLS

- Clojure/Script
- Java
- Python
- AWS
- ElasticSearch
- Natural language processing
- Machine learning
- Semantic web
- Artificial intelligence
- Databases
- ReST/GraphQL